# Muhammad Zaeem Chaudhary

zaeemcodes.net | chaudharycoding@gmail.com | (413) 466 0906 | Linkedin | GitHub

**Programming Languages**: Python, Java, TypeScript, JavaScript, GraphQL, SQL, Rust, C++
**AI Platforms:** Azure AI Foundry, Akamai Conductor, Agent Orchestration, Temporal Workflows, LLM Evaluation, Hugging Face
**Frameworks**: React, React Native, Next.js, Node.js, Express.js, Spring Boot, Flask, TensorFlow, Scikit-learn
**Tools**: AWS, Amazon S3, Kubernetes, Docker, Harness, Git, RESTful APIs, Postman, Snowflake, MongoDB, PostgreSQL

## Education

**University of Massachusetts Amherst** **GPA 3.9** - *Cum Laude* **Graduation: May 2026**
**BS Computer Science** | *Undergraduate Achievement Award* - **Highest undergraduate honor**

## Experience

**Marriott International** **Jun 2026 – Present**
*AI Engineer* Bethesda, MD

- Built agentic search over **multi-source** unstructured data, serving **200K+** users with **governance**, **cost** controls, and AI **safeguards**.
- **Owned** an internal AI tool **end-to-end**, aligning **business** requirements with **engineering execution** to drive platform integration.
- Partnered with **vendors** to introduce scalable **agentic** systems to Marriott's infrastructure, transforming **hospitality** data use cases.
- Scaled the AI **platform team** from **2 to 10,** supporting **hiring**, onboarding, mentorship, and **day-to-day** project delivery.

**Microsoft** **Jan 2026 – Feb 2026**
*Software Engineer Intern* Cambridge, MA

- Worked with the **Health and Life Sciences** team to build a **cloud-based** AI assistant that automates diagnostic **radiology** workflows.
- Built an AI radiology assistant using FastAPI, GPT-5.1, and **spaCy** to automate findings extraction and reduce report time by **80%**.
- Engineered **DuckDB** vector search to map findings to **2,150+** codes, eliminating manual lookup in medical documentation.

**Marriott International** **Jun 2025 – Aug 2025**
*Software Engineer Intern* Bethesda, MD

- Built a **internal chatbot** in Microsoft Power Apps that converts **natural language** to **SQL** via **GPT-4**, built on a Power BI semantic layer.
- Executed auto-generated SQL across **7** domains in **Snowflake** with RBAC and audit logs, returning results with sub-15-second latency.
- Delivered actionable, SQL-transparent data insights with **86%** accuracy, logging lineage and metadata in **Azure** for auditability.

**Snapchat (Snap Inc.)** **Mar 2024 – Apr 2024**
*Software Developer Extern* Remote, CA

- Designed a soccer-themed **AR** lens using Snap's **Lens Studio** and **3D modeling**, certified by Snap Inc.'s Head of Entertainment.
- Launched a soccer-themed Snapchat lens, compatible with **iOS** and **Android**, gathering views from **100+** countries.

**Neftwerk Inc.** **May 2024 – July 2024**
*Data Science Intern* Remote, PA

- Developed **Attio** dashboards and implemented **REST API** filtering, improving client data accuracy and reducing retrieval time by **30%**.
- Engineered **Python** automation for Excel-to-CSV pipeline, leveraging **Git** version control to optimize Attio dashboard data processing.

**Center for Intelligent Information Retrieval** | Research Poster **Sep 2024 – May 2025**
*AI and LLM Research Intern* | *Professor Hamed Zamani* Amherst, MA

- Built an **LLM** personalization platform using **AWS** and **vLLM**; rotated **OpenRouter** models like **Claude 3.5** to analyze user preferences.

**University of Massachusetts Amherst** | Research Poster **Jun 2024 – Aug 2024**
*ML Research Intern* | *June Jeong (PhD)* Amherst, MA

- Optimized a **CNN** for sign-language recognition in **Keras**, achieving a **97% F1** score with data augmentation and batch normalization.

## Projects

**Hack(H)er413 Hackathon Winner: "Best Use of Software"** | Github Code | Demo | *Next.js, Tailwind, Firebase, Gemini.ai*

- Developed an **AI-driven app** for customized interview preparation and organization, enhancing users' interview preparation by **35%**.
- Built an interactive dashboard with a **speech analysis** feature, tailored feedback, and scoring, streamlining user experience by **48%.**

**Lume** | GitHub Code |*Python, Flask, YOLOv8, OpenCV, JavaScript, HTML, CSS*

- Developed a multi-point wildfire detection system using **open-source AI models** to distinguish smoke from fire swiftly.
- Integrated **Twilio** to auto-dispatch alerts; used color space transformations to reduce false detections and increase **precision** by **15%**.